

Deliverables: Dual-Panel Activity & Transport-Layer Protocol Visualizer

1. Project Links

GitHub Repository:

<https://github.com/subhrajyotisaha/NetViz>

Live Application (Railway):

<https://netviz.up.railway.app/>

Google Drive Folder:

<https://drive.google.com/drive/folders/1li-rLlg8MhqExr8c08nTccIEz8EooKE2>

2. Project Overview

NetViz is an AI-assisted network protocol simulator and dual-panel web dashboard designed to bridge the gap between user actions and the underlying application-layer and transport-layer protocols. The application allows users to perform three common activities (Browsing, Mail, and Streaming) and simultaneously visualize the resulting network protocol exchanges (DNS, HTTP, SMTP, TCP, UDP) in real time.

3. Key Features

- **Dual-Panel Layout:** A split-screen interface where user actions are logged on the left, while real-time protocol sequences are animated on the right.
- **Layer Toggle:** A clear interface allowing users to seamlessly switch between high-level Application Layer views and low-level Transport Layer states in parallel.
- **Sequence Diagram Timeline:** An automated, animated vertical timeline that constructs a sequence diagram as the protocols execute, allowing users to step forward, backward, or replay.
- **Detailed Protocol Payloads:** Realistic and highly detailed packet payloads, including full HTTP headers, DNS query flags, and TCP Seq/Ack/Window Size parameters.
- **Color-Coded Execution:** Traffic is uniquely color-coded by protocol (DNS/UDP is Blue, HTTP is Yellow, SMTP is Green, TCP is Red) to allow for instant visual categorization.

4. Technology Stack

- **Frontend Structure:** Pure HTML5 for semantic structure.
- **Styling & Animations:** Tailwind CSS via CDN for modern utility classes, combined with custom CSS keyframes for complex timeline animations.
- **Interaction & Logic:** Vanilla JavaScript (ES6+), utilizing event-driven state machines to govern the synchronization between the activity controls and the visual timeline.
- **AI Platform Used:** Built primarily via Google Antigravity with Gemini 3.1 Pro (High) for rapid architecture planning, CSS animation generation, and accurate TCP protocol flow mockups.

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5. Evaluated Protocol Flows

- **Web Browsing:** Traces a UDP DNS A-record lookup followed by a TCP 3-way handshake, a robust HTTP GET request and standard 200 OK response with PSH/ACK flags, and a TCP teardown (FIN/ACK).
- **Email (SMTP):** Traces a UDP DNS MX-record lookup, followed by a stateful SMTP conversation carried over a continuous TCP stream, including EHLO, STARTTLS encryption, envelope setup, and payload transmission.
- **Media Streaming:** Traces an HTTP-based HLS streaming flow, demonstrating the retrieval of a Master Manifest (.m3u8), a Media Playlist, and sequential video data segments (.ts) transferred via reliable TCP segments.